Chapter 9: Simple Linear Regression

Statistics for Environmental Science

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> *Essentially, all models are wrong, but some are useful.*
>
> George E. P. Box

Back in Chapter 2, we used the covariance between two variables to quantify the strength and direction of a relationship (Pearson's r), fit a best-fit line through a scatterplot, and judged r against a simulated chance window. Chapter 2 gave no formal test of the question every other test in this book has asked:

Could chance alone plausibly have produced our result?

Now that we've learned hypothesis testing, sampling distributions, and the logic of comparing an observed statistic to a null distribution, we can apply that same logic to the regression line and test whether a given slope is evidence of a real relationship.

Regression basics

The equation for a linear regression is one of the most heavily used equations in applied science:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Here y is the dependent variable (DV) and x is the predictor, the independent variable (IV) (Chapter 1). β_1 is the slope, the change in y for each one-unit increase in x. β_0 is the intercept, the value of y when $x = 0$. The intercept and slope are the **regression coefficients**. These subscripted βs have no connection to β, the Type II error rate from Chapter 5.

A statistical **model** is an equation that describes how the DV changes with a predictor, plus a term for the variation the equation leaves out. A linear equation fit to data is one example. Real data do not fall exactly on a line, and the **error term** ε is the part of the DV that the population line does not explain. We never observe β_0, β_1, or ε directly, because they describe the population, and we have only a sample.

Computing the best-fit line by hand

In practice, we let software find the line with the smallest sum of squared residuals (the least-squares line from Chapter 2), but working through one example by hand shows where that line comes from. We write the intercept and slope calculated from a sample as b_0 and b_1, to keep them apart from the population values β_0 and β_1. Here are the formulas for b_0 and b_1, calculated straight from the data:

$$\text{intercept} = b_0 = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$$

$$\text{slope} = b_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Here x and y are the individual scores, n is the number of (x, y) pairs, and $\sum$ means add up that column over all pairs. This is the same line from Figure 2.10 in Chapter 2. To find its slope and intercept, we start with this table:

Table 1: Sums for the slope and intercept formulas

Pair	x	y	x^2	xy
1	1	2	1	2
2	4	5	16	20
3	3	1	9	3
4	6	8	36	48
5	5	6	25	30
6	7	8	49	56
7	8	9	64	72
Sums	34	39	200	231

The table has 7 pairs of x and y scores, plus columns for x^2 and xy (each x score times its matching y score), with column sums at the bottom. These sums are everything the formulas need. From the table:

$$n = 7, \quad \sum x = 34, \quad \sum y = 39, \quad \sum x^2 = 200, \quad \sum xy = 231$$

Substituting:

$$\begin{aligned}
\text{intercept} = b_0 &= \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2} \\
&= \frac{39 \times 200 - 34 \times 231}{7 \times 200 - 34^2} \\
&= \frac{7800 - 7854}{1400 - 1156} \\
&= \frac{-54}{244} = -0.221
\end{aligned}$$

$$\begin{aligned}
\text{slope} = b_1 &= \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \\
&= \frac{7 \times 231 - 34 \times 39}{7 \times 200 - 34^2} \\
&= \frac{1617 - 1326}{1400 - 1156} \\
&= \frac{291}{244} = 1.19
\end{aligned}$$

With b_0 and b_1 in hand, we can compute a residual for any observed point. A residual, as in Chapter 2, is the difference between what we observed and what the line predicts. The symbol $\hat{y}$ (read "y-hat") is the value the line predicts:

$$\begin{aligned}
\hat{y} &= b_0 + b_1 x \\
e &= y - \hat{y}
\end{aligned}$$

For the first row above ($x = 1$, $y = 2$):

$$\begin{aligned}
\hat{y} &= -0.221 + 1.19(1) = 0.969 \\
e &= y - \hat{y} = 2 - 0.969 = 1.031
\end{aligned}$$

The residual is positive, so that point is above the line. Repeating this for every point, then squaring and summing the residuals, gives the sum of squared residuals, the quantity that b_0 and b_1 were chosen to make as small as possible.

The residual e is the sample's version of the error ε. The error ε belongs to the population line, which we never see, and the residual e belongs to the line we fit.

Parameters and sample statistics

The regression coefficients β_0 and β_1 are population parameters.

Symbol	What is it?	Do we know it?
β_0, β_1	True population intercept and slope	No. Estimated from the sample
b_0, b_1	Sample intercept and slope	Yes, calculated by fitting the line to our data

Worked regression example: stream temperature and dissolved oxygen

Warmer water holds less dissolved gas, so ecologists expect dissolved oxygen (DO) to decline as stream temperature rises. This matters for stream health, because low DO can stress or kill fish and invertebrates. Figure 1 lays out the design.

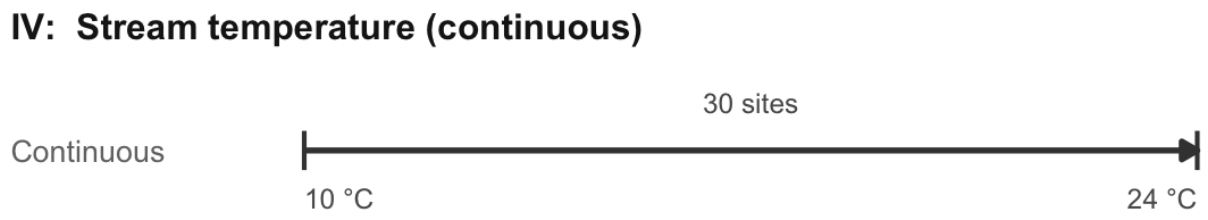


Figure 1: A regression design. Instead of a few groups to sort sites into, every site is somewhere along a continuous range, and every temperature value is treated as its own group. A slope describes how the mean changes across that range.

Up to now every design put units into groups and asked whether the groups differed. Now, with a continuous predictor, every temperature value is treated as its own group. Each of the 30 sites falls somewhere along the range, usually alone at its temperature, and the line gives a predicted mean DO at every temperature, the way a group mean did in earlier chapters. The question becomes how much DO changes per degree rather than whether two groups differ.

The 30 sites in this example are invented. We built them from a population line with $\beta_0 = 13.5$ mg/L and $\beta_1 = -0.28$ mg/L per °C, plus random scatter, so here, unlike in a real study, we know the true slope that b_1 estimates.

The scatterplot in Figure 2 shows a clear downward trend. Just from the graph, we can guess a couple of things:

- Pearson's r should be negative, since DO falls as temperature rises. Pearson's r and the slope b_1 describe the same line in different ways. r measures how closely the points follow a straight line, on a scale from -1 to 1 with no units. b_1 measures how much y changes for each one-unit change in x, in the units of the data (here mg/L per °C).
- Pearson's r should be well short of -1, since the points scatter noticeably around the line.

Regression tests whether a relationship like this one is more than chance by writing it as a model.

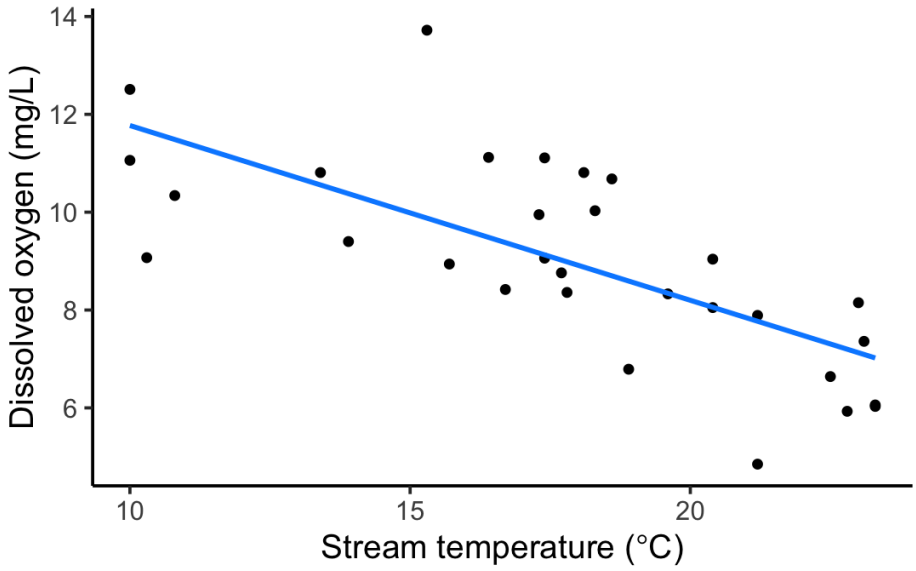


Figure 2: Stream temperature and dissolved oxygen at 30 sample sites, with the fitted regression line.

The regression model

We model dissolved oxygen as a linear function of stream temperature, plus error:

$$DO = \beta_0 + \beta_1(\text{Temperature}) + \varepsilon$$

β_0 is the population intercept, β_1 is the population slope, and ε is the error, the part of DO that temperature alone doesn't explain.

We don't know β_0 and β_1. We estimate them from the sample as b_0 and b_1, using the same least-squares logic from Chapter 2.

In R, this is a single line of code:

```
fit <- lm(DO ~ stream_temp, data = stream_df)
```

In Figure 3, each red vertical segment is one site's residual, the distance from its observed DO to the line. The fitted line is the line that makes the sum of those squared residuals as small as possible.

`summary(fit)` prints the estimate, standard error, t, and p for each coefficient, and `confint(fit)` gives their 95% confidence intervals. The table below combines the two.

```
summary(fit)
confint(fit)
```

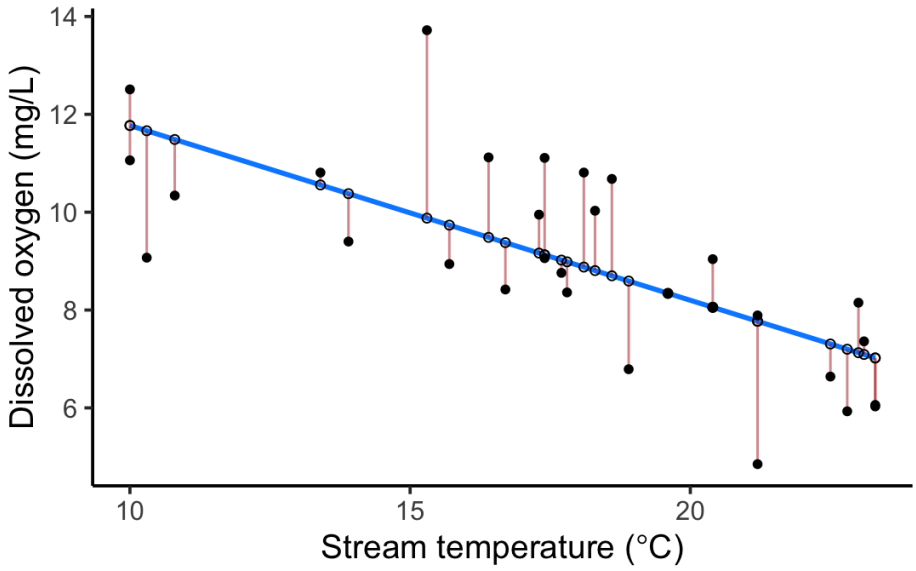


Figure 3: Stream temperature and dissolved oxygen with the fitted line. Vertical red segments are residuals.

Table 2: Coefficient table for DO ~ stream temperature

Term	Estimate	SE	t	p	CI low	CI high
(Intercept)	15.34	1.23	12.47	< 0.001	12.82	17.87
stream_temp	-0.36	0.07	-5.30	< 0.001	-0.50	-0.22

With the results of the regression, we can get an equation for the fitted line:

$\widehat{DO} = 15.34 - 0.36 \times \text{Temperature}$.

The hat over *DO* marks it as a *predicted* value, what the model outputs for a given temperature, as distinct from an actually observed *DO* measurement.

Table 3 says what each column of the coefficient table holds.

Table 3: Columns of the coefficient table, with the names R uses for them.

In this book	In R's `summary()` output	What it is
Estimate	`Estimate`	b_0 and b_1, the fitted intercept and slope (sample values, not β_0 and β_1)
SE	`Std. Error`	How much that estimate would vary from sample to sample
t	`t value`	The estimate divided by its SE
p	Pr(>|t|)	The probability of a t at least this far from 0, in either tail, if the population coefficient were 0
CI low, CI high	`2.5 %` and `97.5 %`, from `confint()`	The bounds of the 95% confidence interval for that coefficient

The intercept ($b_0 = 15.34$) is the model's predicted DO at 0°C. That temperature is outside the range we sampled (10.0 to 23.3°C), so reading the line at 0°C would be extrapolation (Section), and we won't interpret the intercept.

The slope ($b_1 = -0.36$) is the part we care about most. For every 1°C increase in stream temperature, predicted dissolved oxygen drops by about 0.36 mg/L.

Testing the slope

The slope b_1 is a sample estimate. The hypothesis test for the slope asks whether the population slope β_1 could plausibly be zero:

- **Null (H_0):** no linear relationship between stream temperature and dissolved oxygen, $\beta_1 = 0$.
- **Alternative (H_a):** a linear relationship, $\beta_1 \neq 0$.

Like any other statistic, b_1 varies from sample to sample. If we drew a new sample of streams and fit a new line, we'd get a slightly different b_1, just by chance. The test asks whether our b_1 is large relative to that sampling variability, the same signal-over-noise pattern as every test so far:

$$\text{statistic} = \frac{\text{effect}}{\text{error}}.$$

Here "error" means the standard error of b_1, the standard deviation of its sampling distribution. It is a different quantity from the error term ε in the model.

R reports the t-test for the slope in the coefficient table from `summary(fit)`, so there is no separate test to run.

The test statistic divides the estimated slope by its standard error, the SE column in the coefficient table above. R computes the standard error of the slope from the residuals and the spread of temperatures, $SE(b_1) = \sqrt{\dfrac{\sum e^2/(n-2)}{\sum (x - \bar{x})^2}}$. More scatter around the line makes it larger, and a wider spread of temperatures makes it smaller. For the stream data,

$$t = \frac{b_1}{SE(b_1)} = \frac{-0.357}{0.067} = -5.30$$

with $df = n - 2 = 28$ (one degree of freedom spent estimating the intercept, one for the slope).

The p-value comes from a t-distribution. If temperature and DO were truly unrelated in the population, repeating this study many times would still produce a nonzero b_1 just by chance, and a t-statistic that follows a t-distribution with $df = 28$. Figure 4 shows that distribution, with the critical values marking the boundary of the most extreme 5% under H_0, and our observed t marked against it.

Our observed $t = -5.30$ is far past the critical values of about ± 2.05, well outside where 95% of t-values would fall if temperature and DO were unrelated. The p-value is the probability of a t that far out (or farther) in either tail under H_0. Here $p < 0.001$, which is smaller than $\alpha = 0.05$.

We reject H_0 and conclude that stream temperature is linearly related to dissolved oxygen.

The last two columns of the coefficient table give the 95% confidence interval for the slope, 95% CI [-0.50, -0.22] mg/L per °C. Because this interval does not contain 0, it agrees with the hypothesis test.

This is the same agreement you saw for z in Chapter 4 and for t in Chapter 6. Here the critical value is $t_{crit} = 2.05$ at $df = 28$. The test rejects H_0 when the observed b_1 is more than t_{crit} standard errors from 0,

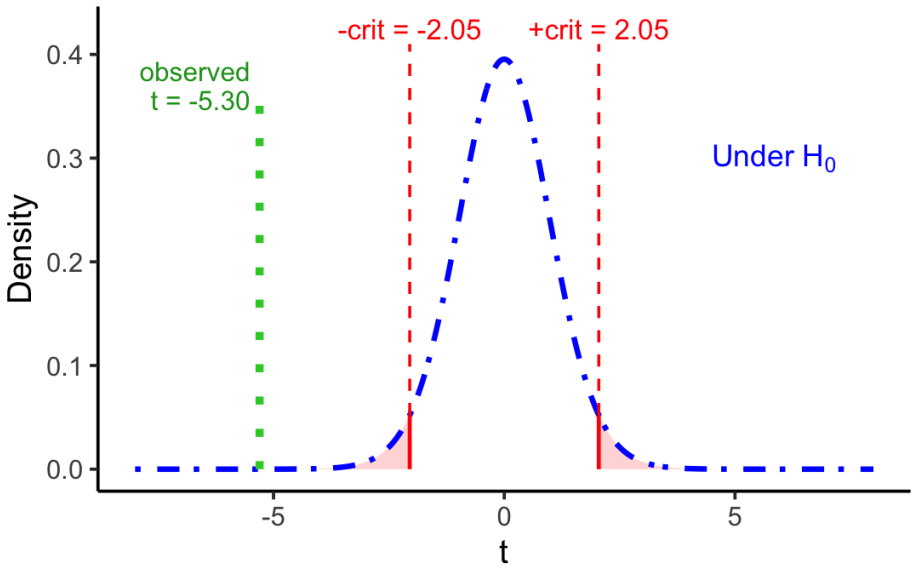


Figure 4: The null distribution of t for the slope test at $df = 28$ (blue dot-dash). The shaded tails beyond the red critical values are the rejection region at $\alpha = 0.05$. The lime-green dotted line marks our observed t, far into the left tail.

and the 95% interval contains every slope within t_{crit} standard errors of b_1. So 0 falls outside the interval exactly when the two-tailed test rejects H_0 at $\alpha = 0.05$.

The interval also contains $\beta_1 = -0.28$, the true slope we built into the data. A 95% interval catches the true value in about 95% of samples, and this sample is one of them.

Partitioning variation: regression as ANOVA

Besides testing the slope, we want to know how much of the variation in DO the line explains. We answer that the same way as one-way ANOVA in Chapter 8, by splitting the total variation into a piece the model explains and a piece it leaves over.

Figure 5 shows the three pieces for the stream data. Each site's distance from mean DO (gray, left panel) splits into two pieces, the distance from the mean to the line (middle), which the line explains, and the distance from the line to the point (right), the residual.

SS_{Total} is the sum of the squared distances from each point to mean DO. $SS_{\text{Regression}}$ is the sum of the squared distances from the line to mean DO. SS_{Residual} is the sum of the squared residuals, the distances from each point to the line. For each site, the two shorter segments add up to the long one. After squaring each segment and summing over all 30 sites, the totals still add up, as they did in Chapter 8:

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Residual}}, \qquad 122.44 = 61.35 + 61.10$$

The pieces line up with Chapter 8's. $SS_{\text{Regression}}$ is the variation the predictor explains, the role SS_{Between} played for group means. SS_{Residual} is what's left over around the line, the role SS_{Within} played around each group's mean. Treating every temperature value as its own group means that the line supplies a predicted mean at each value of x, in place of one mean per group.

R-squared (R^2) is the proportion of the total variation explained by the line:

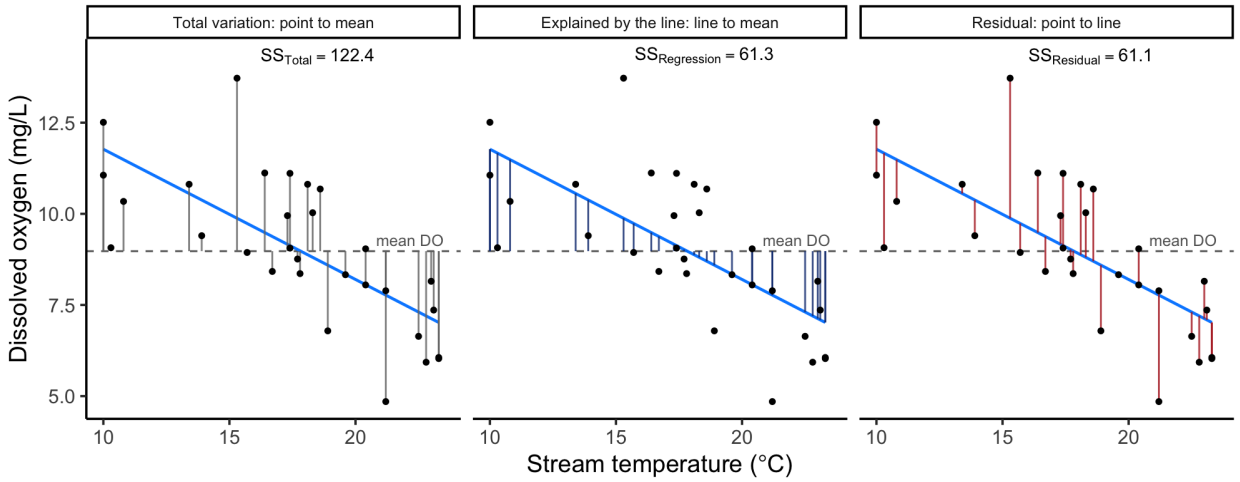


Figure 5: Partitioning variation in dissolved oxygen. Left: each site's distance from mean DO (dashed line). Middle: the part of that distance explained by the line. Right: what is left over, the residuals. Squaring and summing each set of segments gives the three sums of squares.

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}} = \frac{61.35}{122.44} = 0.50$$

It runs from 0 to 1, and it is the same quantity as η^2 from Chapter 8. For simple regression with one predictor, R^2 is also Pearson's r from Chapter 2, squared: $(-0.708)^2 = 0.501$.

The regression ANOVA table. Dividing each sum of squares by its degrees of freedom gives mean squares, and their ratio is F, exactly as in Chapter 8. The regression row uses 1 degree of freedom, one for the slope. The residual row uses $n - 2$, since the line estimated two things, an intercept and a slope.

```
anova(fit)
```

Table 4: Regression ANOVA table for DO ~ stream temperature.

Source	df	SS	MS	F	p
stream_temp	1	61.35	61.35	28.12	< 0.001
Residuals	28	61.10	2.18		

$$F = \frac{MS_{\text{Regression}}}{MS_{\text{Residual}}} = \frac{SS_{\text{Regression}}/1}{SS_{\text{Residual}}/28} = \frac{61.35/1}{61.10/28} = \frac{61.35}{2.182} = 28.12$$

With one predictor, F and the slope's t test the same thing: $F = t^2 = (-5.30)^2 = 28.12$, with the same p. F is the form that generalizes to models with more than one predictor, which is why software reports both.

Here $R^2 = 0.50$, so stream temperature explains about 50% of the variation we observed in dissolved oxygen across sites. The remaining 50% comes from everything else the model doesn't capture (discharge, shading, groundwater input, and so on), and it shows up as the residuals.

Consider both significance and effect size

As with every test in this book, we report an effect size alongside the significance test. Testing the slope asks whether β_1 differs from 0. R^2, regression's effect size, says how much of the variation the line explains. The two can disagree in either direction.

A sample of thousands of observations, common in remote sensing and other big-data environmental work, can make a trivial effect significant, because more observations shrink the standard error until almost any nonzero slope clears $p < 0.05$. A small sample can fail the other way and miss a real, sizeable effect because its standard error is too large.

Figure 6 shows three of these combinations, using invented data: a strong relationship with a non-significant slope (underpowered, Chapter 5), a weak relationship with a significant slope (a large sample), and a relationship that is both strong and significant. In the left panel, six points give $R^2 = 0.63$ but $p = 0.058$. In the middle panel, 5,000 points give $p = 0.004$ for a line that explains 0.2% of the variation. We built a true slope of 0.05 into that panel. With x running from 0 to 10, the line rises $0.05 \times 10 = 0.5$ units across the whole range.

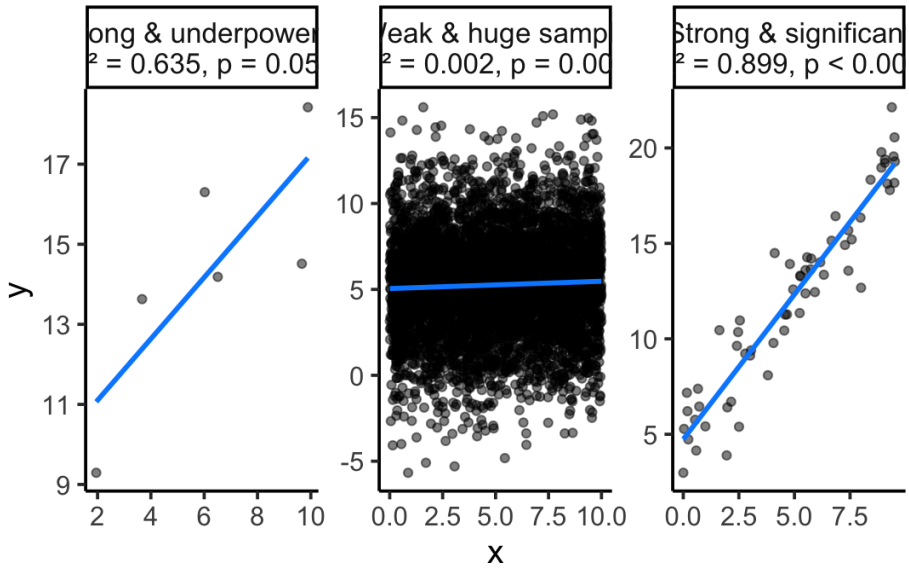


Figure 6: R-squared and the slope's p-value can disagree. Left: a strong relationship (high R-squared) that isn't statistically significant. Middle: a weak relationship (low R-squared) that is significant anyway. Right: strong and significant together.

> Warning: How big is a "good" R^2?
>
> It depends on the field. In studies of human behavior, where many unmeasured factors shape any outcome, 0.20 is a strong result. Environmental systems tend toward more direct physical relationships, so 0.5 to 0.8 for a well-chosen single predictor isn't unusual. An R^2 that looks *too* good, say 0.98 from a single predictor in a noisy field system, more often points to a mistake in the analysis, such as a predictor that is computed from the DV, than to a real result.

Checking assumptions

Simple linear regression has three test-specific assumptions, plus the universal independence assumption from Chapter 6. Table 5 lists all four with the check this chapter uses.

Table 5: Assumptions of simple linear regression and how to check them.

Assumption	What it means	How to check	What a violation looks like
Linearity	The true relationship between x and y is a straight line.	Residuals-versus-fitted plot	A curve, such as a U-shape
Independence	Each observation's error is unrelated to the others'.	The study design	Sites along the same stream, or repeated measurements over time
Constant variance (homoscedasticity)	The spread of residuals is about the same across the whole range of x, the regression version of Chapter 8's homogeneity of variance.	Residuals-versus-fitted plot	A funnel that widens or narrows
Normal residuals	The residuals are approximately normally distributed.	Q-Q plot and Shapiro-Wilk	Points bending away from the Q-Q diagonal

When the true relationship is curved

> You can do linear regression without thinking about whether the phenomenon you're modeling is actually close to linear. But you shouldn't.
> Jordan Ellenberg, *How Not to Be Wrong*

A straight line may be the wrong shape for your data. A residuals-versus-fitted plot puts each point's fitted value $\hat{y}$, the value the line predicts for it, on the x-axis and its residual on the y-axis. It checks two of the assumptions above at once. A curve means linearity is violated, and a funnel shape means constant variance is violated.

Figure 7, for our stream temperature data, shows residuals scattered evenly above and below zero, with no curve and no funnel shape.

To see what a violation looks like, we fit a straight line to a second data set, 14 days of invented avian influenza case counts. We built the counts from an exponential curve, $6e^{0.28 \times \text{day}}$, which grows by about 32% each day, plus random scatter, so we know the true shape that a straight line will miss. The design has the same structure as the stream example, with time as the continuous predictor.

A straight line fit to these 14 days gives $R^2 = 0.81$ and $p < 0.001$, which is higher than the 0.5 to 0.8 that is common for a single predictor in environmental work, and significant. Neither number alone suggests a problem, but the plot does. The fitted line starts at about -40 cases on day 1, and a count of cases can't be negative.

Check the residuals. Figure 10 shows the residuals-versus-fitted plot for the outbreak model.

The residuals form a U. The residuals are positive early and late in the outbreak, so the observed counts are above the line there and the line underpredicts. In the middle they are negative, so the line overpredicts.

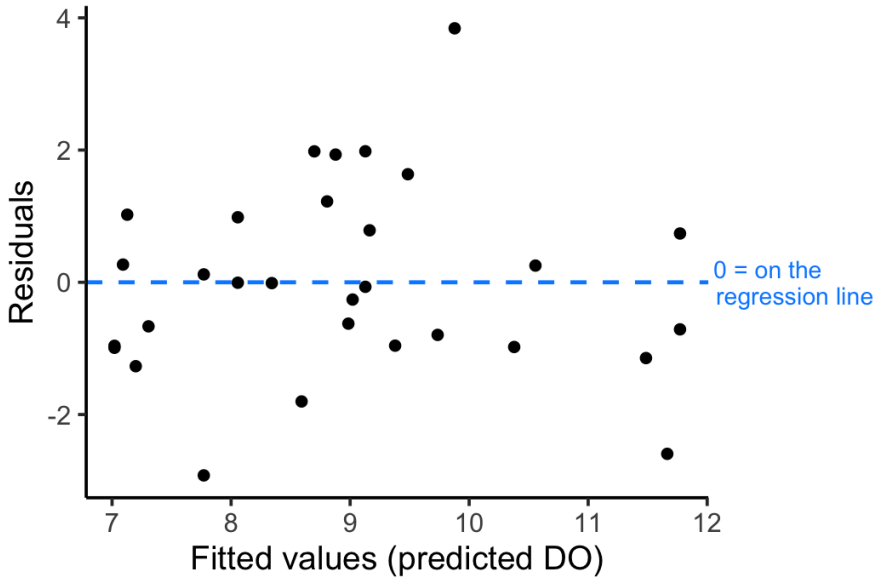


Figure 7: Residuals versus fitted values for the stream temperature model. A random scatter around zero, with no funnel shape or curve, supports linearity and constant variance.

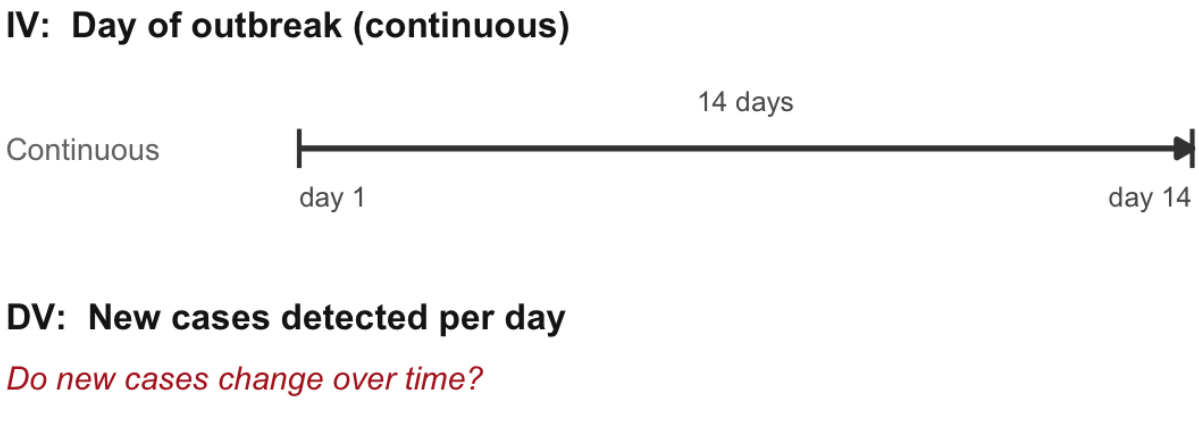


Figure 8: The outbreak data as a design.

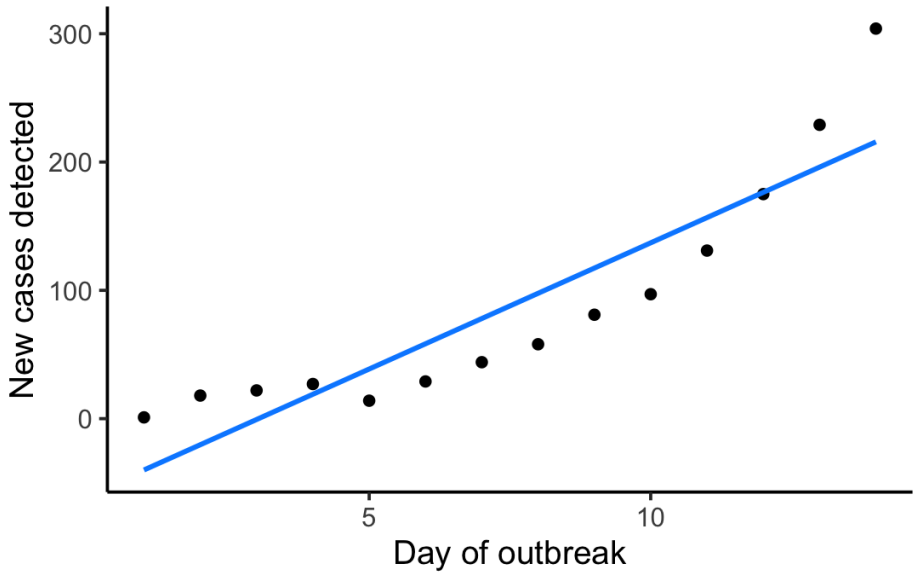


Figure 9: New avian influenza cases per day, days 1-14 of an outbreak, with a fitted straight line.

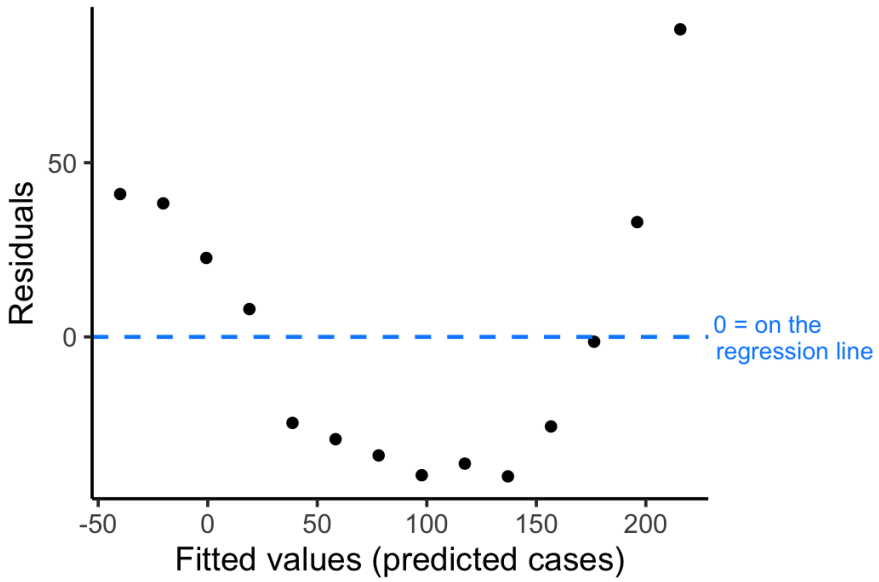


Figure 10: Residuals versus fitted values for the linear outbreak model. The residuals form a U-shape instead of a random scatter, which happens when a straight line is fit to a curved relationship.

If a residual plot curves instead of scattering randomly, a straight line is the wrong model for the data.

Fixes typically include:

- a different model (often adding an x^2 term, which lets the fitted line bend)
- a transformation, re-expressing a variable on a different scale so a straight line fits it (log-transforming a variable that grows multiplicatively, like the outbreak data above, is the most common case)
- or a nonlinear model entirely, beyond what this course covers

Checking normality

For normality, we check the residuals two ways: visually, with a Q-Q plot, and formally, with the Shapiro-Wilk test (Chapter 6).

A **Q-Q plot** sorts the residuals and compares them to where a perfectly normal distribution would place values at the same percentiles. If the residuals are normal, the points fall close to the diagonal line. A systematic curve or S-shape means they don't.

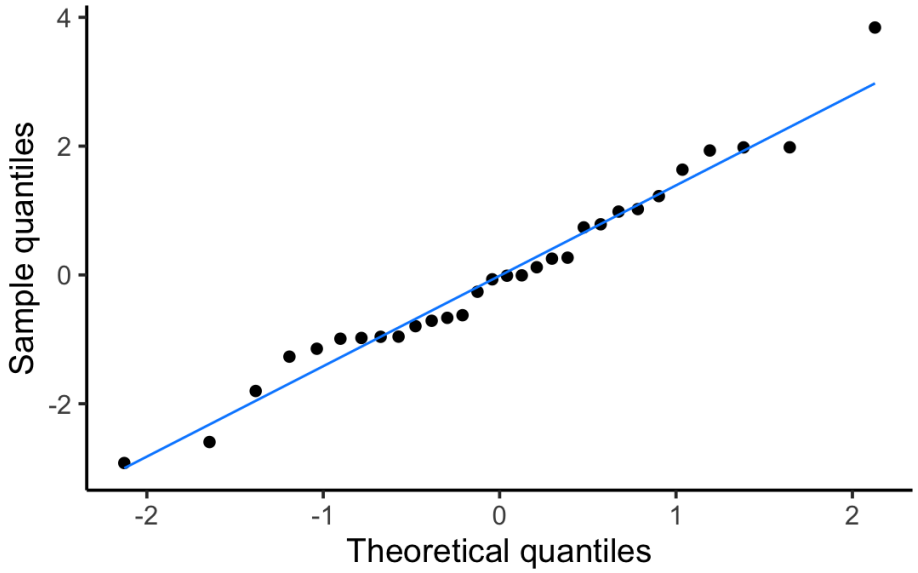


Figure 11: Normal Q-Q plot of the stream temperature model's residuals. Points close to the diagonal support normality.

Figure 11 looks close to straight, consistent with normal residuals. Shapiro-Wilk asks the same question formally. Its H_0 is that the residuals come from a normal distribution, so a small p-value is evidence against normality.

```
shapiro.test(resid(fit))
```

Table 6: Shapiro-Wilk test on the regression residuals

Statistic	p
0.97	0.663

Shapiro-Wilk does not reject normality here ($p = 0.663$). As with any formal normality test, treat this as one piece of evidence alongside the Q-Q plot and residual plot. Shapiro-Wilk has the least power to detect real departures when the sample is small, which is also when the visual checks matter most.

Now apply the same two normality checks to the avian influenza model, the one we already know is badly nonlinear.

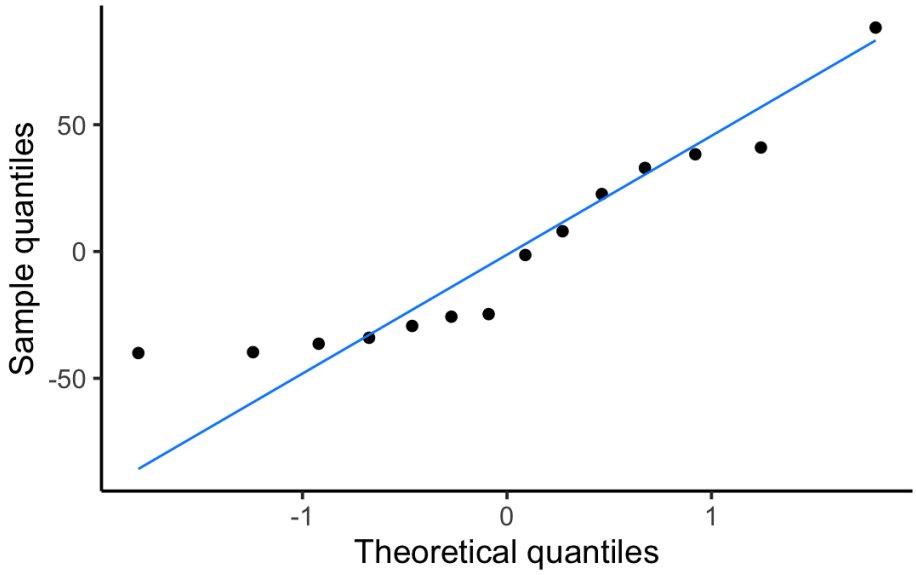


Figure 12: Normal Q-Q plot of the outbreak model's residuals. The lowest residuals bend away from the diagonal, but not by enough for Shapiro-Wilk to reject normality.

The Q-Q plot bends away from the diagonal at the low end, but Shapiro-Wilk does not reject normality ($p = 0.060$). On a model we already know is the wrong shape, the normality checks give at most borderline evidence of a problem. They check *normality of the residuals*, which is a different assumption from linearity, and a model can badly violate linearity while its residuals look only mildly non-normal.

! Results reporting: regression

We report a regression result the same way as any other test: the slope, the test statistic with its degrees of freedom, p, the slope's confidence interval, and R^2 as the effect size.

> Dissolved oxygen declined significantly with stream temperature, $b_1 = -0.36$ mg/L per °C, $t(28) = -5.30$, $p < 0.001$, 95% CI [-0.50, -0.22] mg/L per °C, $R^2 = 0.50$.

Table 7 collects the slope test in simple linear regression on one card.

Table 7: Test card: simple linear regression.

Purpose	Is one continuous variable linearly related to another? One predictor, one outcome
Hypotheses	$H_0 : \beta_1 = 0$ $H_a : \beta_1 \neq 0$
Assumptions	Linearity Constant variance of the residuals across x Approximately normal residuals

Test statistic	$t = b_1 / SE(b_1)$
	$df = n - 2$
	Equivalently, $F = t^2$ with 1 and $n - 2$ df
Report	b_1 in its units, with the 95% CI for the slope
	$t(df)$ and p
	R^2

Using the model: prediction

Once the slope test shows a significant linear relationship, the fitted equation lets us predict DO at any temperature inside the range we sampled. Plugging temperature into the fitted line directly gives the prediction: at 20°C,

$$\widehat{DO} = b_0 + b_1(20) = 15.34 + (-0.36)(20) = 8.20$$

R gives the same number, along with a confidence interval. `predict()` takes the fitted model and a small data frame holding the new temperature. The column name, `stream_temp`, has to match the predictor's name in the model.

```
predict(fit, newdata = data.frame(stream_temp = 20), interval = "confidence")
```

Table 8: Predicted dissolved oxygen at 20°C, with 95% confidence interval

Stream temperature	Predicted	CI low	CI high
20	8.2	7.57	8.83

At 20°C, the model predicts DO of 8.20 mg/L, 95% CI [7.57, 8.83]. Because 20°C falls inside the observed range of 10.0 to 23.3°C, this prediction is interpolation. **Interpolation** is predicting y at an x value inside the range of x we observed, which is what regression is good for.

This confidence interval is for the population mean DO of all streams at 20°C, the value the model predicts at $x = 20$, $\beta_0 + 20\beta_1$. It is a different interval from the ones in the coefficient table, which give a range of plausible values for each parameter, β_0 and β_1. A single stream at 20°C will scatter around that mean by about as much as the points scatter around the line, so its DO can fall well outside this interval.

Extrapolation is predicting y at an x value outside the range of x we observed. Predicting DO at, say, 35°C would be extrapolation. The fitted line gives 2.8 mg/L at 35°C, a number with no data behind it. Nothing in the data tells us the relationship stays linear (or even stays a relationship at all) beyond the range we actually observed, and the further outside that range, the less trustworthy the prediction. Figure 13 shows both zones on the same line.

Regression is used for extrapolation all the time, and often reasonably. A well-understood physical process can justify projecting a line beyond the data we collected. Extrapolating a straight line is only safe if the true relationship stays straight past the range we observed.

Figure 14 shows four very different relationships, exponential growth, a sigmoid, a hump-shaped curve, and a truly straight line, each with the same short observed window highlighted in blue. In each panel a straight line (dashed red) is fit to that window and extended. Once you see the full curve (gray), you can see how badly linear extrapolation beyond the range of observed data would perform in each case.

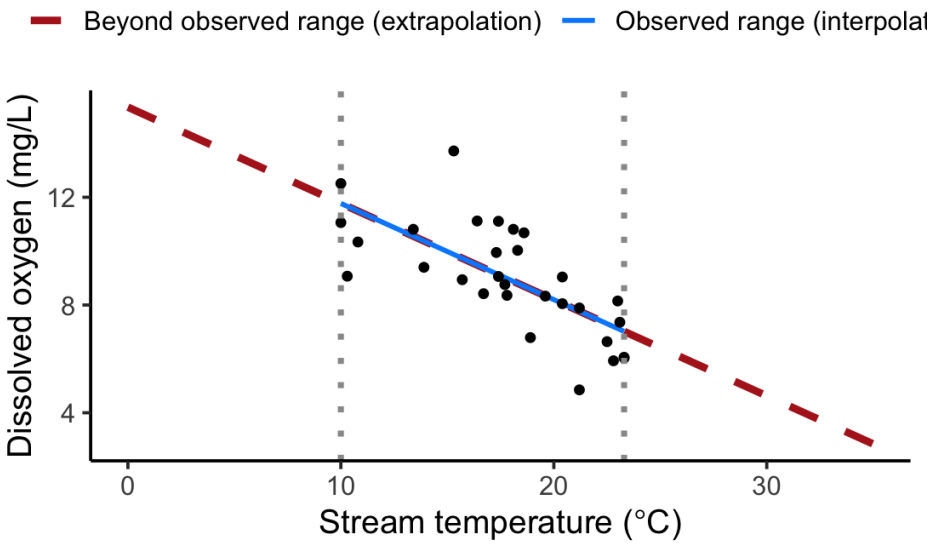


Figure 13: The fitted line within the observed temperature range (solid blue, interpolation) and extended beyond it in both directions (dashed red, extrapolation). Dotted gray lines mark the lowest and highest observed temperatures, 10.0 and 23.3°C.

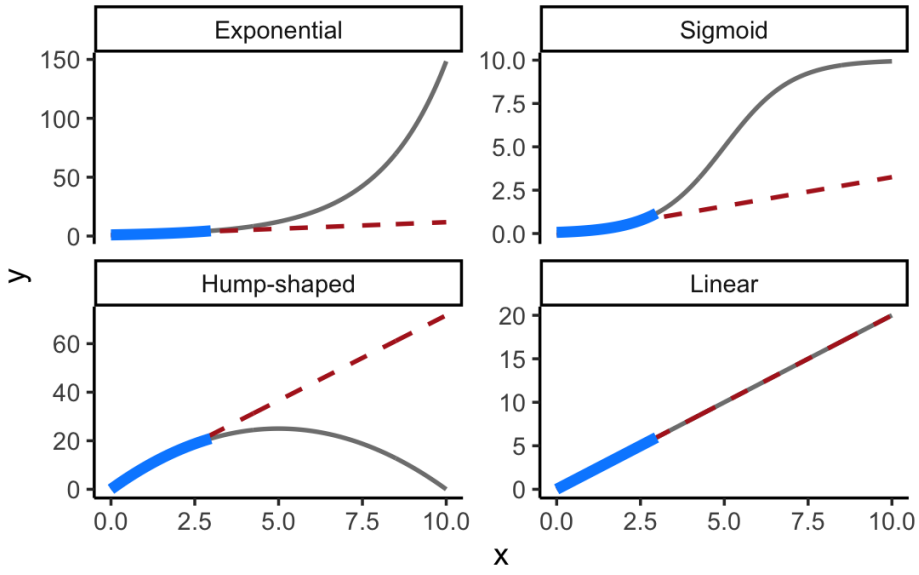


Figure 14: Four different true relationships that all look locally linear over a short observed window (blue). A line fit to each window (dashed red) matches it closely inside the window, but only the linear case stays correct once the full curve (gray) is revealed.

Suppose officials use the linear fitted line from the outbreak example above to project avian influenza cases at day 30, well past the 14 days observed.

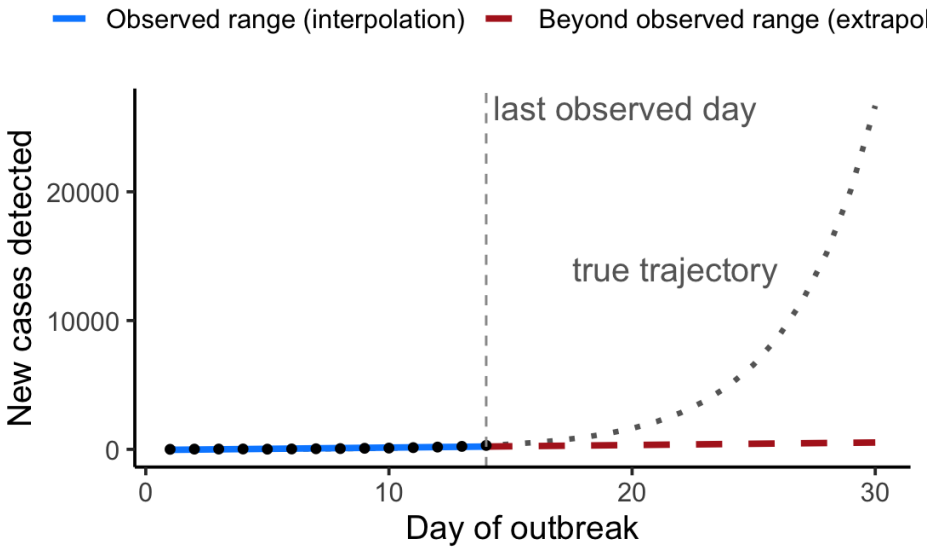


Figure 15: The linear fit extended (dashed) past the observed range, against the true exponential trajectory (dotted gray). The straight line badly underestimates how large the outbreak actually becomes.

In Figure 15, the dashed line keeps rising slowly past day 14, while the true exponential curve rises far above it.

The linear model predicts about 530 cases on day 30. On the exponential curve the data were built from, the count is closer to 26,682, roughly 50 times higher.

Planning surveillance and containment staffing off the linear forecast would leave the response badly under-resourced for the outbreak that actually arrives.

This problem calls for a different model. More data or a better-fitting straight line would not fix it. Fitting a curve or a transformed model is beyond this course. You can still recognize when a straight line is the wrong model by plotting the data and the residuals.

Chapter Summary

Why it matters. Chapter 2 drew a best-fit line through a scatterplot. This chapter tests that line's slope, measures how much variation it explains with R^2, and uses it to predict new values. Chapter 11 adds a categorical factor to this same line.

Core ideas

- **The slope can be tested.** Chapter 2 compared r to a simulated chance window but gave no formal test with a p-value. Fitting a line turns that relationship into a model whose slope we can test: $H_0 : \beta_1 = 0$ says there is no linear relationship. The test statistic is the slope divided by its standard error, $t = b_1/SE(b_1)$, with $df = n - 2$.

- **Regression partitions variation the way ANOVA does.** SS_{Total} splits into $SS_{\text{Regression}}$ (Chapter 8's between) and SS_{Residual} (Chapter 8's within), and $F = MS_{\text{Regression}}/MS_{\text{Residual}}$, which equals t^2 with one predictor. R^2, the proportion of variation explained by the line, is the effect size. A relationship can be real (p small) and still explain very little, which is why you report both.
- **Look at your data, then look at your residuals.** The test-specific assumptions are linearity, constant variance, and roughly normal residuals, on top of independence. A residuals-versus-fitted plot can show a violation that R^2, the slope test, and Shapiro-Wilk all miss, as the outbreak example showed.
- **Predict inside the observed range, and extrapolate beyond it only with good reason.** Interpolating within the observed range of x is what regression is for. Extrapolating beyond it is sometimes justified, but only when you have good reason to believe the relationship stays the same shape past the range you observed.
- **A good R^2 can still be the wrong model.** The outbreak example fit a straight line to exponential growth and got $R^2 = 0.81$, a residuals-versus-fitted plot that clearly curved, and an extrapolated forecast about 26,000 cases short of the exponential curve the data came from. A normality check on the same residuals gives only a borderline signal, because it tests a different assumption.